MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

3120 SCREEN BUILDING

DESIGN CALCULATION OF BAG FILTER & DUCT SUPPORT FOUNDATION

AS BUILT

MD-216-3120-EG-CV-CAL-2021

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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2007	Saudi Code for Concrete Structures

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
RL	:	Reduced Level
C.L.	:	Centre Line
CJ	:	Construction Joint
FDN	:	Foundation
TOC	:	Top of concrete
FGL	:	Finished Ground Level
FFL	:	Finished Floor Level
BOF	:	Bottom of footing
TYP	:	Typical
TOC	:	Top of concrete
IP	:	Insert Plate
RCC	:	Reinforced Cement Concrete
MKD.	:	Marked
THK.	:	Thick

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1. INTENT

The document is prepared for foundation design calculation of Deducting duct support, Bag filter and its access staircase.
This bag filter is near Screen building and TP-2.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

This document is prepared for foundation design of Deducting duct support, Bag filter and its access staircase. All the loads applied to the structure for analysis & design followed Civil -Structural Design criteria and SBC 301:2007. The column and beam reinforcement design have been performed in Staad-Pro Software. The design of combined footings have been performed in MS Excel.
Total foundation scheme has been modelled and analysed using FEM analysis in Staad-Pro Software.

4. CONCRETE STRUCTURE

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

4.2 Materials

Concrete Strength of 28 days
· Foundation and Structural Members: 35 MPa
· Grade slab and pavement: 30 MPa
· Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade
· Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
· Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundations for Structures

Foundation shall be Isolated and combined Shallow Concrete foundation. The design calculations are prepared as per recommendation of standard SBC-304 Clause 13.3.

4.4 Minimum Concrete Cover

Items	Minimum Cover(mm)
Foundation (Side & Bottom)	75
Foundation (Top)	50
Column & Pedestal	50
Beam	50
Slab	40

4.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 500 kN/m2 at 3.5m depth below FGL (ie. RL 909.000m) refer Table 4 - BH-17,18,19,25,29 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.6 Reference

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

5 INPUT LOADS FROM SUPERSTRUCTURE

5.1 Support Reaction from superstructure

For load calculation and support reaction of TP-2 superstructure, refer Document No: MD-216-3120-EG-ST-CAL-1005
For load calculation and support reaction of Duct supporting superstructure, refer Document No: MD-216-3120-EG-ST-CAL-2022
For support reaction of Bag filter superstructure, refer Drawing No: MD-216-3100-EG-ME-DED-V509-2002

	Bag Filter Structure		
	Due to Wind (in KN)	Due to Seismeic (in KN)	Guiding LC
Total load along +X direction	613	383	Wind
Total load along -X direction	612	383	Wind
Total load along +Z direction	808	271	Wind
Total load along -Z direction	806	271	Wind

As total base shear due to seismic force in each direction is much lesser than the wind load in respective direction. So, in foundation design load combination considering seismic loads are not considered.

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5.2 LOAD COMBINATIONS

5.2.1 Load Combination for Base pressure calculation

(As per Section 2.4 of SBC 301 -2007)

Load Cases	Reference
LOAD COMB 201 1.0 (DL + SDL1 + LL)	(Eq. 2.4.1-1 of SBC 301)
LOAD COMB 202 1.0 (DL + SDL2 + LL)	
LOAD COMB 203 1.0 (DL+ SDL1) + 0.75 WL(+X) + 0.75 LL	(Eq. 2.4.1-6 of SBC 301)
LOAD COMB 204 1.0 (DL+ SDL1) + 0.75 WL(-X) + 0.75 LL	
LOAD COMB 205 1.0 (DL+ SDL1) + 0.75 WL(+Z) + 0.75 LL	
LOAD COMB 206 1.0 (DL+ SDL1) + 0.75 WL(-Z) + 0.75 LL	
LOAD COMB 207 1.0 (DL+ SDL2) + 0.75 WL(+X) + 0.75 LL	
LOAD COMB 208 1.0 (DL+ SDL2) + 0.75 WL(-X) + 0.75 LL	
LOAD COMB 209 1.0 (DL+ SDL2) + 0.75 WL(+Z) + 0.75 LL	
LOAD COMB 210 1.0 (DL+ SDL2) + 0.75 WL(-Z) + 0.75 LL	
LOAD COMB 211 0.6 (DL+ SDL1) + 1.0 WL(+X)	(Eq. 2.4.1-7 of SBC 301)
LOAD COMB 212 0.6 (DL+ SDL1) + 1.0 WL(-X)	
LOAD COMB 213 0.6 (DL+ SDL1) + 1.0 WL(+Z)	
LOAD COMB 214 0.6 (DL+ SDL1)+ 1.0 WL(-Z)	
LOAD COMB 215 0.6 (DL+ SDL2) + 1.0 WL(+X)	
LOAD COMB 216 0.6 (DL+ SDL2) + 1.0 WL(-X)	
LOAD COMB 217 0.6 (DL+ SDL2) + 1.0 WL(+Z)	
LOAD COMB 218 0.6 (DL+ SDL2)+ 1.0 WL(-Z)	
LOAD COMB 219 1.0 (DL +SDL1 + TL1) + 1.0 LL	(Eq. 2.4.1-2 of SBC 301)
LOAD COMB 220 1.0 (DL + SDL1 + TL2) + 1.0 LL	
LOAD COMB 221 1.0 (DL + SDL2 + TL1) + 1.0 LL	
LOAD COMB 222 1.0 (DL + SDL2 + TL2) + 1.0 LL	

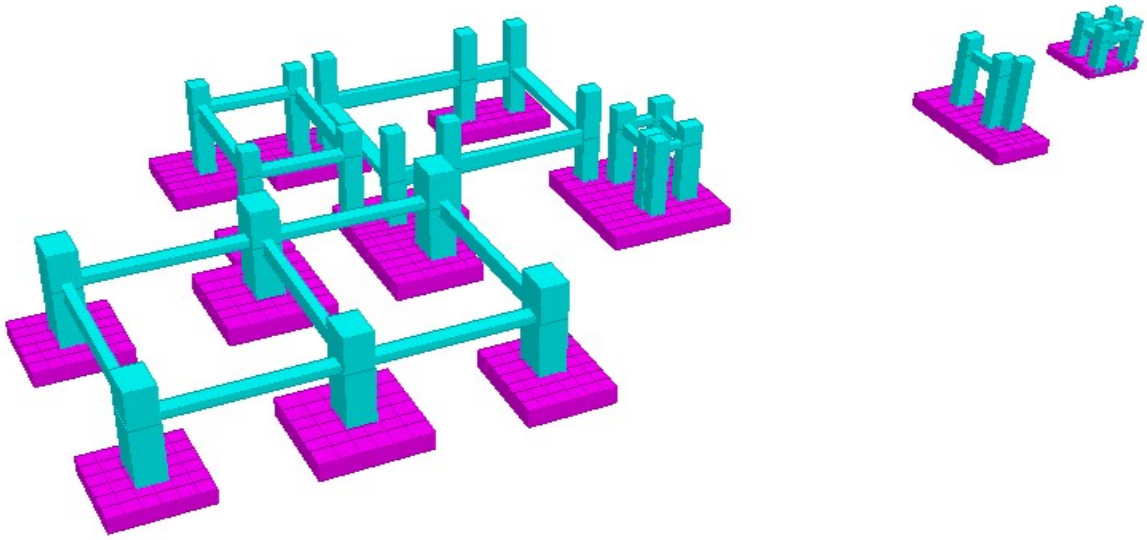
5.2.2 Load Combination for Concrete design

Load Case	Reference
LOAD COMB 301 1.4 (DL + SDL1)	(Eq. 2.3.2-1 of SBC 301)
LOAD COMB 302 1.4 (DL + SDL2)	
LOAD COMB 303 1.4 (DL +SDL1 + TL1) + 1.7 LL	(Eq. 2.3.2-2 of SBC 301)
LOAD COMB 304 1.4 (DL + SDL1 + TL2) + 1.7 LL	
LOAD COMB 305 1.4 (DL + SDL2 + TL1) + 1.7 LL	
LOAD COMB 306 1.4 (DL + SDL2 + TL2) + 1.7 LL	
LOAD COMB 307 1.2 (DL+ SDL1) + 1.6 WL(+X) + 0.5 LL	(Eq. 2.3.2-4 of SBC 301)
LOAD COMB 308 1.2 (DL+ SDL1) + 1.6 WL(-X) + 0.5 LL	
LOAD COMB 309 1.2 (DL+ SDL1) + 1.6 WL(+Z) + 0.5 LL	
LOAD COMB 310 1.2 (DL+ SDL1) + 1.6 WL(-Z) + 0.5 LL	
LOAD COMB 311 1.2 (DL+ SDL2) + 1.6 WL(+X) + 0.5 LL	
LOAD COMB 312 1.2 (DL+ SDL2) + 1.6 WL(-X) + 0.5 LL	
LOAD COMB 313 1.2 (DL+ SDL2) + 1.6 WL(+Z) + 0.5 LL	
LOAD COMB 314 1.2 (DL+ SDL2) + 1.6 WL(-Z) + 0.5 LL	
LOAD COMB 315 0.9 (DL+ SDL1) + 1.6 WL(+X)	(Eq. 2.3.2-6 of SBC 301)
LOAD COMB 316 0.9 (DL+ SDL1) + 1.6 WL(-X)	
LOAD COMB 317 0.9 (DL+ SDL1) + 1.6 WL(+Z)	
LOAD COMB 318 0.9 (DL+ SDL1)+ 1.6 WL(-Z)	
LOAD COMB 319 0.9 (DL+ SDL2) + 1.6 WL(+X)	
LOAD COMB 320 0.9 (DL+ SDL2) + 1.6 WL(-X)	
LOAD COMB 321 0.9 (DL+ SDL2) + 1.6 WL(+Z)	
LOAD COMB 322 0.9 (DL+ SDL2)+ 1.6 WL(-Z)	

Eq. 2.3.2-3 of SBC 301 - 2007 has not been considered as it would not produce worst load combination in the absence of Roof Live Load & Rain Load

6 GEOMETRY

6.1 Geometry of Foundation arrangement



Isometric view of total foundation arrangement (From Staad)

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7. DESIGN OF CONCRETE COMPONENTS

All foundations have been designed using worksheet in accordance to SBC-304.

All beams & columns have been designed in Staad-Pro by ACI 318M.

For Design Results, Refer Attachemnt 5 - Staad Output file.

7.1 Design Summary of Column

7.1.1 Column Reinforcement Summary

Column mark	Width	Depth	Reinforcement	Stirrup	Remarks	Numbers
PD1	450	450	8 No.- D20	3 Nos D10 @ 250c/c	Pedestals for duct support	4
PD2	500	700	12 No.- D20	3 Nos D10 @ 250c/c	Pedestals for duct support	2
PD3	500	700	12 No.- D20	3 Nos D10 @ 250c/c	Pedestals for duct support	4
PC1	550	550	12 No.- D20	3 Nos D10 @ 250c/c	Pedestals for conv support	1
PC2	500	550	12 No.- D20	3 Nos D10 @ 250c/c	Pedestals for conv support	1
PT1	600	600	12 No.- D25	3 Nos D10 @ 250c/c	Pedestals for Transfer Point mkd. TP-2	4
PT2	600	600	12 No.- D25	3 Nos D10 @ 250c/c	Pedestals for Transfer Point mkd. TP-2	2
PB1	900	1200	26 No.- D25	7 Nos D10 @ 250c/c	Pedestals for Bag filter support	6
PS1	600	600	12 No.- D20	3 Nos D10 @ 250c/c	Pedestals for duct support	4

7.2 Design Summary of Beam

7.2.1 Beam Reinforcement Summary

Beam mark	Width (mm)	Depth (mm)	Top Reinforcement	Bottom Reinforcement	Stirrup	Numbers
TB1	300	300	3 Nos D12	3 Nos D12	2L D10 STRPS @ 175c/c	4
TB2	300	400	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	5
TB3	300	600	3 Nos D20	3 Nos D20	2L D10 STRPS @ 250c/c	7
TB4	300	500	3 Nos D20	3 Nos D20	2L D10 STRPS @ 250c/c	8

7.3 Design of Foundation

7.3.1 Foundation marking, Max. Base pressure and Reinforcement Summary

Structure Name	Foundation mark	Length (Along vertical) (mm)	Width (Along horizontal) (mm)	Thickness (mm)	Max pressure (kN/m^2)	Top reinforcement	Bottom reinforcement	F.O.S.	Numbers
Bag filter stair support	F1	2000	2000	400	122	D12 @ 200c/c B/W	D16 @ 200c/c B/W	**0.24**	1
Bag filter supp.	F2	3000	3000	600	132	D16 @ 150c/c B/W	D20 @ 150c/c B/W	**0.26**	2
Bag filter supp.	F3	3500	3500	700	213	D16 @ 150c/c B/W	D20 @ 150c/c B/W	**0.43**	2
Bag filter supp.	F4	2900	3500	700	213	D16 @ 150c/c B/W	D20 @ 150c/c B/W	**0.43**	1
Bag filter stair support	F5	3000	3000	600	132	D16 @ 150c/c B/W	D20 @ 150c/c B/W	**0.26**	1
TR1 of Duct supp.	CF1	2500	2500	400	81	D12 @ 200c/c B/W	D16 @ 200c/c B/W	**0.16**	1
TR2 of Duct supp. & Conv. Trestle supp.	CF2	2400	5400	600	132	D16 @ 150c/c B/W	D20 @ 150c/c B/W	**0.26**	1
TR3 of Duct supp. & Conv. Trestle supp. & TP2 pedestal support	CF3	4500	5000	600	132	D16 @ 150c/c B/W	D20 @ 150c/c B/W	**0.26**	1
TP2 pedestal support	CF4	5000	3800	600	132	D16 @ 150c/c B/W	D20 @ 150c/c B/W	**0.26**	1
TP2 pedestal support & Bag filter stair support	CF5	3500	2900	600	132	D16 @ 150c/c B/W	D20 @ 150c/c B/W	**0.26**	1
Bag filter supp, Bag filter stair support & TP2 pedestal support	CF6	4400	5500	700	213	D16 @ 150c/c B/W	D20 @ 150c/c B/W	**0.43**	1

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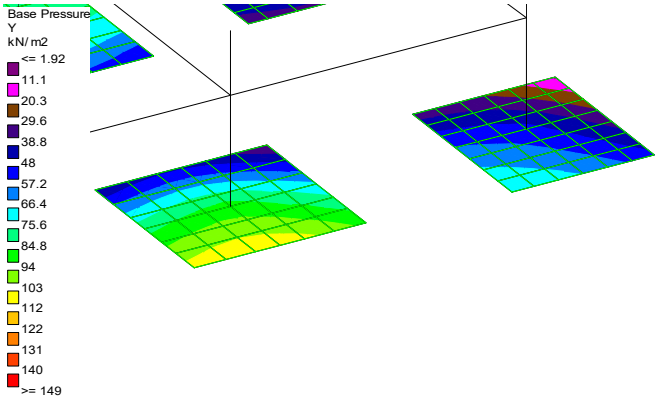
7.3.2 Design of footing mkd. F3, F4 & CF6

For Calculation of Base pressure and design, Refer STAAD File: Screen Duct Support_Foundation_Final.std

Staad Supoort Nodes are 5 & 6



Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +912.750m)
Finished Ground level	F.G.L	=	0.0	m	(Corresponds to RL +912.250m)
Bottom of foundation	B.O.F	=	-3.50	m	(Corresponds to RL +910.000m)
Top of pedestal (staad support)	T.O.F	=	0.35	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	3.500	m	
Width of footing	Lx	=	3.500	m	
Thickness of footing	D	=	0.70	m	
Depth of pedestal	Bx	=	0.60	m	
Width of pedestal	Bz	=	0.60	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	563	KN/m^2	
Soil load on footing in staad		=	50.4	KN/m^2	staad input

1) Check for Gross Base Pressure :

Maximum Base Pressure		=	213.9	KN/m^2	Load Case 205
Minimum Base Pressure		=	-12.7	KN/m^2	Load Case 215

2) Design of foundation Raft :

Slab Thickness =	700	mm	Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Width =	1000	mm	Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Clear Cover =	75	mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Compressive Strength of Concrete , fc' =	35	N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

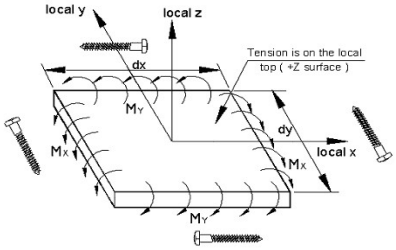
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	126.463	kN-m	Corresponding Mxy	32.475	kN-m	*(Plate No. - 731 , L/C No. - 320)*
Min. Mx	-337.011	kN-m	Corresponding Mxy	24.995	kN-m	*(Plate No. - 650 , L/C No. - 308)*
Max. My	208.668	kN-m	Corresponding Mxy	-18.766	kN-m	*(Plate No. - 851 , L/C No. - 321)*
Min. My	-393.856	kN-m	Corresponding Mxy	30.430	kN-m	*(Plate No. - 851 , L/C No. - 310)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	126.463 + 32.475 =	158.938	kN-m
My1 =	My + abs(Mxy) =	208.668 + 18.766 =	227.434	kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	126.463 + 5.054 =	131.517	kN-m
My2 =	My+abs(Mxy^2/Mx) =	208.668 + 6.808 =	215.476	kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd =	158.938	kN-m					
Myd =	227.434	kN-m					

ALONG LOCAL X - DIRECTION :

Mu =	158.938	kN-m						
Φ*Kn =	Mu/bd^2					deff. =	617	mm
Mu/bd^2 =	158.938X10^6/(1000X617^2) =	0.418	N/mm^2	<	6.9	N/mm^2		
Pt =	0.0011	Reqd. Ast =	0.0011X1000X617/100 =	6.79	cm^2			
Provided	16	mm @	150	C/C	(Provide Ast =	13.4	cm^2)	
					Total Provide Ast =	13.4	cm^2	O.K.
					Total Provide Pt =	0.0022		

ALONG LOCAL Y - DIRECTION :

Mu =	227.434	kN-m						
Φ*Kn =	Mu/bd^2					deff. =	601	mm
Mu/bd^2 =	227.434X10^6/(1000X601^2) =	0.597	N/mm^2	<	6.9	N/mm^2		
Pt =	0.0016	Reqd. Ast =	0.0016X1000X617/100 =	9.62	cm^2			
Provided	16	mm @	150	C/C	(Provide Ast =	13.4	cm^2)	
					Total Provide Ast =	13.4	cm^2	O.K.
					Total Provide Pt =	0.0022		

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	03-09-2020	14-10-2020	31-10-2023				Date	03-09-2020	14-10-2020	31-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -337.011 - 24.995 = -362.006 kN-m
My1 = My - abs(Mxy) = -393.856 - 30.43 = -424.286 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -337.011 - 1.586 = -338.597 kN-m
My2 = My-abs(Mxy^2/Mx) = -393.856 - 2.748 = -391.108 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -362.006 kN-m
Myd = -424.286 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 362.006 kN-m
Φ*Kn = Mu/bd^2 deff. = 615 mm
Mu/bd^2 = 362.006X10^6/(1000X615^2) = 0.957 N/mm^2 < 6.9 N/mm^2
Pt = 0.0026 Reqd. Ast = 0.0026X1000X615/100 = 15.99 cm^2
Provided 20 mm @ 150 C/C (Provide Ast = 20.94)cm^2

Total Provide Ast = 20.94 cm^2 O.K.
Total Provide Pt = 0.0034

ALONG LOCAL Y - DIRECTION :

Mu = 424.286 kN-m
Φ*Kn = Mu/bd^2 deff. = 595 mm
Mu/bd^2 = 424.286X10^6/(1000X595^2) = 1.198 N/mm^2 < 6.9 N/mm^2
Pt = 0.0032 Reqd. Ast = 0.0032X1000X595/100 = 19.04 cm^2
Provided 20 mm @ 150 C/C (Provide Ast = 20.94)cm^2

Total Provide Ast = 20.94 cm^2 O.K.
Total Provide Pt = 0.0035

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = 0.628 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 738 , L/C No. - 307)*
Vu = 386.22 kN

Provided Pt = 0.0034
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0034X386.22X10^3X615/362.006/10^6)X1000X615/10^3
= 605.466 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0034)X1000X615/10^3
= 617.689 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X615/10^3
= 1055.133 kN
HENCE, Vc = 605.466 kN
Φt*Vc = 0.75X605.466 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 454.1 kN > 386.22T , Hence O.K.

ALONG LOCAL Y - DIRECTION :

SQy = 0.538 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 607 , L/C No. - 310)*
Vu = 320.11 kN

Pt provided = 0.0035
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0035X320.11X10^3X595/424.286/10^3)X1000X595/10^3
= 579.103 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0035)X1000X595/10^3
= 598.613 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X595/10^3
= 1020.82 kN
HENCE, Vc = 579.103 kN
Φt*Vc = 0.75X579.103 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 434.327 kN > 320.11T , Hence O.K.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

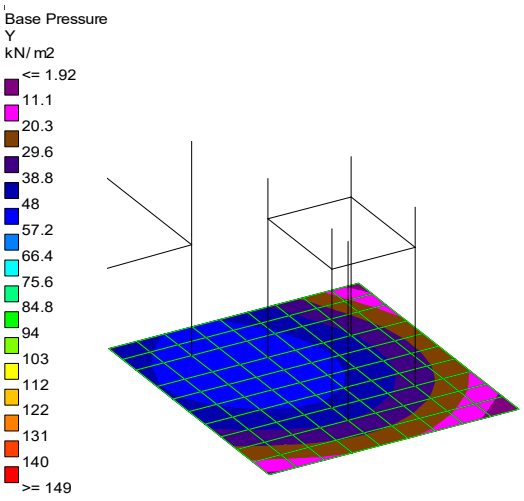
Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	03-09-2020	14-10-2020	31-10-2023	Date	03-09-2020	14-10-2020	31-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL	Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

7.3.3 Design of footing mkd. F2, F5, CF2,CF3, CF4 & CF5

For Calculation of Base pressure and design, Refer STAAD File: Screen Duct Support_Foundation_Final.std

Staad Supoort Nodes are 5 & 6

Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +912.750m)
Finished Ground level	F.G.L	=	0.0	m	(Corresponds to RL +912.250m)
Bottom of foundation	B.O.F	=	-3.50	m	(Corresponds to RL +910.000m)
Top of pedestal (staad support)	T.O.F	=	0.35	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	3.500	m	
Width of footing	Lx	=	3.500	m	
Thickness of footing	D	=	0.60	m	
Depth of pedestal	Bx	=	0.60	m	
Width of pedestal	Bz	=	0.60	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	563	KN/m^2	
Soil load on footing in staad		=	52.2	KN/m^2	staad input



1) Check for Gross Base Pressure :

Maximum Base Pressure	=	132.0	KN/m^2	Load Case 205
Minimum Base Pressure	=	-9.0	KN/m^2	Load Case 215

2) Design of foundation Raft :

Slab Thickness =	600	mm	Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Width =	1000	mm	Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Clear Cover =	75	mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Compressive Strength of Concrete , fc' =	35	N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

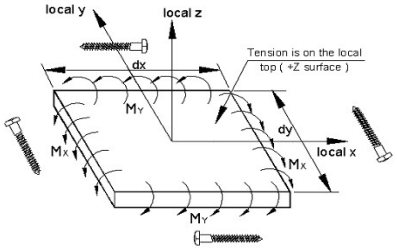
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	44.337	kN-m	Corresponding Mxy	-45.458	kN-m	*(Plate No. - 411 , L/C No. - 317)*
Min. Mx	-265.174	kN-m	Corresponding Mxy	-30.688	kN-m	*(Plate No. - 371 , L/C No. - 310)*
Max. My	91.521	kN-m	Corresponding Mxy	-8.489	kN-m	*(Plate No. - 446 , L/C No. - 317)*
Min. My	-241.465	kN-m	Corresponding Mxy	-71.997	kN-m	*(Plate No. - 531 , L/C No. - 314)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	44.337 + 45.458 =	89.795	kN-m
My1 =	My + abs(Mxy) =	91.521 + 8.489 =	100.010	kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	44.337 + 22.579 =	66.916	kN-m
My2 =	My+abs(Mxy^2/Mx) =	91.521 + 8.097 =	99.618	kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 89.795 kN-m
Myd = 100.01 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 89.795 kN-m
Φ*Kn = Mu/bd^2 deff. = 517 mm
Mu/bd^2 = 89.795X10^6/(1000X517^2) = 0.336 N/mm^2 < 6.9 N/mm^2
Pt = 0.0009 Reqd. Ast = 0.0009X1000X517/100 = 4.65 cm^2
Provided 16 mm @ 150 C/C (Provide Ast = 13.4)cm^2

Total Provide Ast = 13.4 cm^2 O.K.
Total Provide Pt = 0.0026

ALONG LOCAL Y - DIRECTION :

Mu = 100.01 kN-m
Φ*Kn = Mu/bd^2 deff. = 501 mm
Mu/bd^2 = 100.01X10^6/(1000X501^2) = 0.374 N/mm^2 < 6.9 N/mm^2
Pt = 0.0010 Reqd. Ast = 0.001X1000X517/100 = 5.01 cm^2
Provided 16 mm @ 150 C/C (Provide Ast = 13.4)cm^2

Total Provide Ast = 13.4 cm^2 O.K.
Total Provide Pt = 0.0027

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	03-09-2020	14-10-2020	31-10-2023				Date	03-09-2020	14-10-2020	31-10-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -265.174 - 30.688 = -295.862 kN-m
My1 = My - abs(Mxy) = -241.465 - 71.997 = -313.462 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -265.174 - 3.9 = -269.074 kN-m
My2 = My-abs(Mxy^2/Mx) = -241.465 - 19.548 = -221.917 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -295.862 kN-m
Myd = -313.462 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 295.862 kN-m
Φ*Kn = Mu/bd^2 deff. = 515 mm
Mu/bd^2 = 295.862X10^6/(1000X515^2) = 1.116 N/mm^2 < 6.9 N/mm^2
Pt = 0.0030 Reqd. Ast = 0.003X1000X515/100 = 15.45 cm^2
Provided 20 mm @ 150 C/C (Provide Ast = 20.94)cm^2

Total Provide Ast = 20.94 cm^2 O.K.
Total Provide Pt = 0.0041

ALONG LOCAL Y - DIRECTION :

Mu = 313.462 kN-m
Φ*Kn = Mu/bd^2 deff. = 495 mm
Mu/bd^2 = 313.462X10^6/(1000X495^2) = 1.279 N/mm^2 < 6.9 N/mm^2
Pt = 0.0035 Reqd. Ast = 0.0035X1000X495/100 = 17.33 cm^2
Provided 20 mm @ 150 C/C (Provide Ast = 20.94)cm^2

Total Provide Ast = 20.94 cm^2 O.K.
Total Provide Pt = 0.0042

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.615 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 421 , L/C No. - 309)*
Vu = 316.73 kN

Provided Pt = 0.0041
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0041X316.725X10^3X515/295.862/10^6)X1000X515/10^3
= 507.275 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0041)X1000X515/10^3
= 523.38 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X515/10^3
= 883.567 kN
HENCE, Vc = 507.275 kN
Φt*Vc = 0.75X507.275 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 380.456 kN **> 316.725T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = -0.658 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 410 , L/C No. - 313)*
Vu = 325.71 kN

Pt provided = 0.0042
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0042X325.71X10^3X495/313.462/10^3)X1000X495/10^3
= 486.732 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0042)X1000X495/10^3
= 503.897 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X495/10^3
= 849.253 kN
HENCE, Vc = 486.732 kN
Φt*Vc = 0.75X486.732 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 365.049 kN **> 325.71T , Hence O.K.**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	03-09-2020	14-10-2020	31-10-2023	Date	03-09-2020	14-10-2020	31-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL	Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

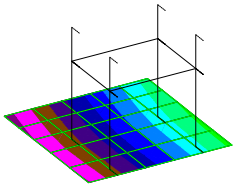
7.3.4 Design of footing mkd. F1 & CF1

For Calculation of Base pressure and design, Refer STAAD File: Screen Duct Support_Foundation_Final.std

Staad Supoort Nodes are 5 & 6

Finished Floor level	F.F.L	=	0.3	m	(Corresponds to RL +912.750m)
Finished Ground level	F.G.L	=	0.0	m	(Corresponds to RL +912.250m)
Bottom of foundation	B.O.F	=	-3.50	m	(Corresponds to RL +910.000m)
Top of pedestal (staad support)	T.O.F	=	0.35	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	3.000	m	
Width of footing	Lx	=	3.000	m	
Thickness of footing	D	=	0.40	m	
Depth of pedestal	Bx	=	0.60	m	
Width of pedestal	Bz	=	0.60	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	563	KN/m^2	
Soil load on footing in staad		=	55.8	KN/m^2	staad input

Base Pressure
Y
kN/ m2
<= 1.92
11.1
20.3
29.6
38.8
48
57.2
66.4
75.6
84.8
94
103
112
122
131
140
>= 149



1) Check for Gross Base Pressure :

Maximum Base Pressure	=	123.0	KN/m^2	Load Case 205
Minimum Base Pressure	=	-10.0	KN/m^2	Load Case 215

2) Design of foundation Raft :

Slab Thickness =	400	mm	Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Width =	1000	mm	Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Clear Cover =	75	mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Compressive Strength of Concrete , fc' =	35	N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

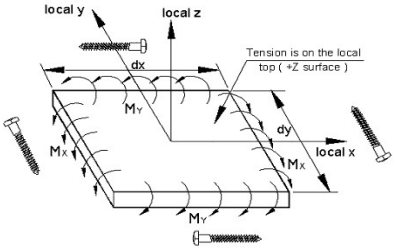
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	21.107	kN-m	Corresponding Mxy	-7.468	kN-m	*(Plate No. - 277 , L/C No. - 319)*
Min. Mx	-68.793	kN-m	Corresponding Mxy	26.787	kN-m	*(Plate No. - 277 , L/C No. - 308)*
Max. My	16.707	kN-m	Corresponding Mxy	-7.468	kN-m	*(Plate No. - 277 , L/C No. - 319)*
Min. My	-58.168	kN-m	Corresponding Mxy	26.787	kN-m	*(Plate No. - 277 , L/C No. - 308)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	21.107 + 7.468 =	28.575	kN-m
My1 =	My + abs(Mxy) =	16.707 + 7.468 =	24.175	kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	21.107 + 3.338 =	24.445	kN-m
My2 =	My+abs(Mxy^2/Mx) =	16.707 + 2.642 =	19.349	kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 28.575 kN-m
Myd = 24.175 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 28.575 kN-m
Φ*Kn = Mu/bd^2 deff. = 319 mm
Mu/bd^2 = 28.575X10^6/(1000X319^2) = 0.281 N/mm^2 < 6.9 N/mm^2
Pt = 0.0007 Reqd. Ast = 0.0007X1000X319/100 = 2.23 cm^2
Provided 12 mm @ 200 C/C (Provide Ast = 5.65)cm^2

Total Provide Ast = 5.65 cm^2 O.K.
Total Provide Pt = 0.0018

ALONG LOCAL Y - DIRECTION :

Mu = 24.175 kN-m
Φ*Kn = Mu/bd^2 deff. = 307 mm
Mu/bd^2 = 24.175X10^6/(1000X307^2) = 0.238 N/mm^2 < 6.9 N/mm^2
Pt = 0.0006 Reqd. Ast = 0.0006X1000X319/100 = 1.84 cm^2
Provided 12 mm @ 200 C/C (Provide Ast = 5.65)cm^2

Total Provide Ast = 5.65 cm^2 O.K.
Total Provide Pt = 0.0018

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	03-09-2020	14-10-2020	31-10-2023				Date	03-09-2020	14-10-2020	31-10-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -68.793 - 26.787 = -95.58 kN-m
My1 = My - abs(Mxy) = -58.168 - 26.787 = -84.955 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -68.793 - 12.336 = -81.129 kN-m
My2 = My-abs(Mxy^2/Mx) = -58.168 - 10.43 = -47.738 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -95.58 kN-m
Myd = -84.955 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 95.58 kN-m
Φ*Kn = Mu/bd^2 deff. = 317 mm
Mu/bd^2 = 95.58X10^6/(1000X317^2) = 0.951 N/mm^2 < 6.9 N/mm^2
Pt = 0.0026 Reqd. Ast = 0.0026X1000X317/100 = 8.24 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0032

ALONG LOCAL Y - DIRECTION :

Mu = 84.955 kN-m
Φ*Kn = Mu/bd^2 deff. = 301 mm
Mu/bd^2 = 84.955X10^6/(1000X301^2) = 0.938 N/mm^2 < 6.9 N/mm^2
Pt = 0.0025 Reqd. Ast = 0.0025X1000X301/100 = 7.53 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0033

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.632 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 277 , L/C No. - 308)*
Vu = 200.34 kN

Provided Pt = 0.0032
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0032X200.344X10^3X317/95.58/10^6)X1000X317/10^3
= 311.522 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0032)X1000X317/10^3
= 317.308 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X317/10^3
= 543.865 kN
HENCE, Vc = 311.522 kN
Φt*Vc = 0.75X311.522 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 233.642 kN > 200.344T , Hence O.K.

ALONG LOCAL Y - DIRECTION :

SQy = 0.491 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 277 , L/C No. - 308)*
Vu = 147.791 kN

Pt provided = 0.0033
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0033X147.791X10^3X301/84.955/10^3)X1000X301/10^3
= 293.76 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0033)X1000X301/10^3
= 301.805 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X301/10^3
= 516.415 kN
HENCE, Vc = 293.76 kN
Φt*Vc = 0.75X293.76 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 220.32 kN > 147.791T , Hence O.K.

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	03-09-2020	14-10-2020	31-10-2023				Date	03-09-2020	14-10-2020	31-10-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 1 **MECHANICAL GENERAL ARRANGEMENT DRAWINGS**

DRAWING NUMBER: MD-216-3100-EG-ME-DED-V509-2001
REV: A01

DESCRIPTION OF LOAD CASES	
LOAD CASE	LOAD TYPE
1	DEAD LOAD
2	LIVE LOAD
3	WIND LOAD +X DIRECTION
4	WIND LOAD -X DIRECTION
5	WIND LOAD +Z DIRECTION
6	WIND LOAD -Z DIRECTION
7	SEISMIC LOAD +X DIRECTION
8	SEISMIC LOAD -X DIRECTION
9	SEISMIC LOAD +Z DIRECTION
10	SEISMIC LOAD -Z DIRECTION

COLUMN	LOAD CASE	FX (kN)	FY (kN)	FZ (kN)
A1	1	0	-207	-8
	2	0	-391	-19
	3	4	164	26
	4	-4	-163	-25
	5	0	441	123
	6	0	-441	-123
	7	2	163	7
	8	-2	-164	-7
	9	0	256	45
	10	0	-254	-45

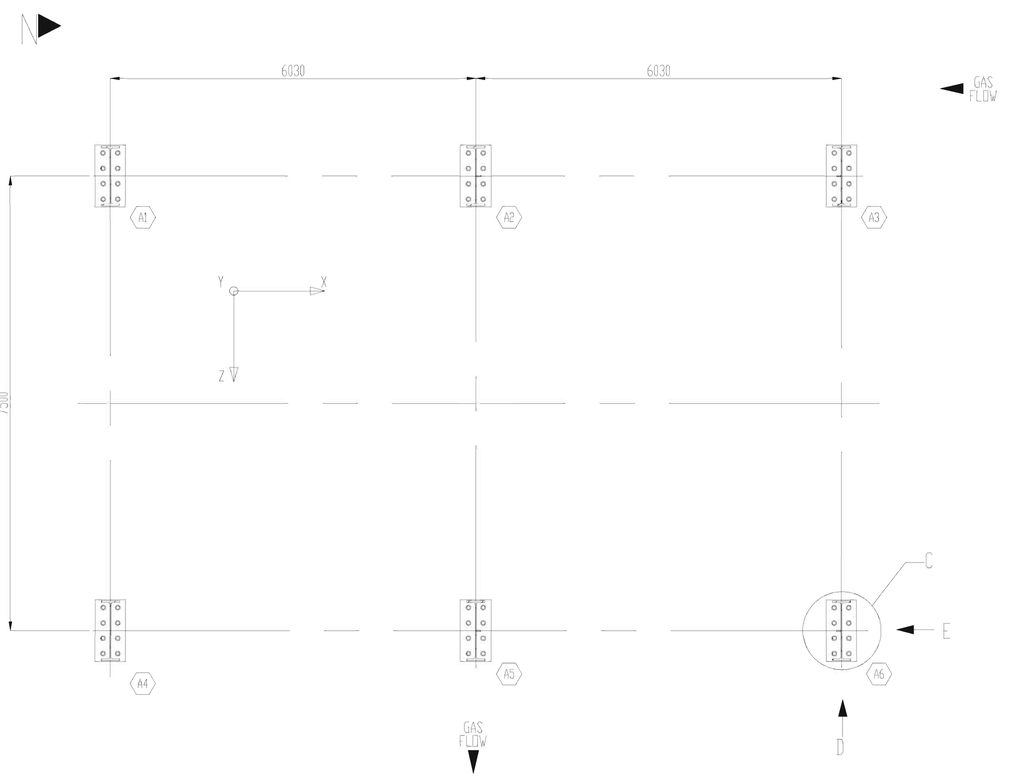
COLUMN	LOAD CASE	FX (kN)	FY (kN)	FZ (kN)
A2	1	0	-347	-14
	2	0	-784	-37
	3	171	278	1
	4	-4	-275	-2
	5	46	684	138
	6	0	-683	-138
	7	189	271	-1
	8	-2	-268	1
	9	13	396	46
	10	0	-374	-46

COLUMN	LOAD CASE	FX (kN)	FY (kN)	FZ (kN)
A3	1	0	-212	-8
	2	0	-395	-19
	3	3	-442	-28
	4	-167	438	26
	5	-2	465	143
	6	-43	-466	-143
	7	1	-433	-6
	8	-188	432	6
	9	-13	255	44
	10	0	-253	-45

COLUMN	LOAD CASE	FX (kN)	FY (kN)	FZ (kN)
A4	1	0	-205	8
	2	0	-377	19
	3	7	105	15
	4	-6	-109	-14
	5	0	-441	123
	6	0	441	-123
	7	2	164	-7
	8	-2	-165	7
	9	0	-255	45
	10	0	256	-45

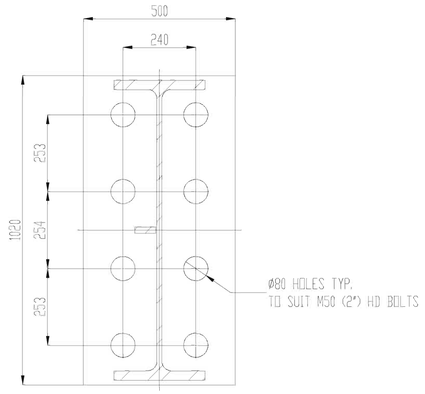
COLUMN	LOAD CASE	FX (kN)	FY (kN)	FZ (kN)
A5	1	0	-343	14
	2	0	-756	37
	3	426	598	3
	4	-6	-595	-3
	5	0	-683	138
	6	44	684	-136
	7	188	264	2
	8	-2	-264	-1
	9	0	-373	46
	10	13	370	-46

COLUMN	LOAD CASE	FX (kN)	FY (kN)	FZ (kN)
A6	1	0	-209	8
	2	0	-378	19
	3	2	-704	-18
	4	-425	705	19
	5	-43	-466	143
	6	0	465	-143
	7	1	-428	6
	8	-187	428	-6
	9	0	-253	45
	10	-13	255	-44

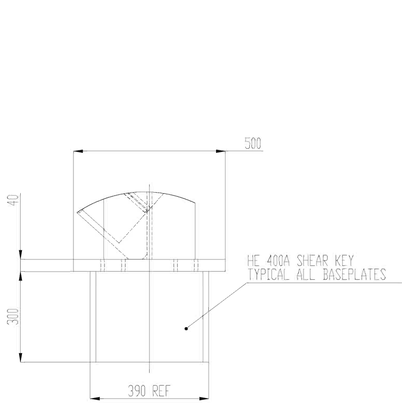


PLAN VIEW AT PA960X5500 FILTER BASEPLATES

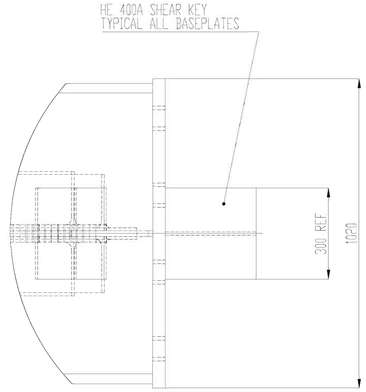
GENERAL NOTES:
ANCHOR BOLTS TO ASTM F1554 SECTION 6.6.1 & TABLE 3
NUTS TO ASTM A563, GRADE A, HEAVY HEX WITH UNC-2B THREADS
WASHERS TO ASTM F436
ALTERNATE USE METRIC EQUIVALENTS



TYPICAL BASEPLATE
DETAIL C
SCALE 1 : 10



VIEW D



VIEW E

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01	15/07/2020	ISSUED FOR INFORMATION		TKS	ABD	SBA	

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DRAWING NUMBER	10701-004	REVISION	C
DRAWN	MDM	DATE	01/05/2020
CHECKED	AMc	DATE	02/07/2020
APPROVED	AMc	DATE	02/07/2020

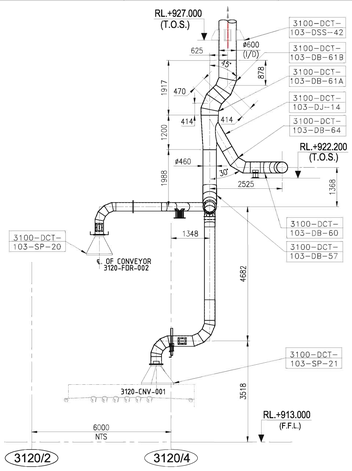
LARSEN & TOUBRO DRAWING NUMBER	O19045_080_U_V_21419_001	REVISION	0
OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__CTB17__31006	DOCUMENT ID	OU100682856
		REVISION	0

DRAWN		
CHECKED	TKS	15/07/2020
ENGINEER	ABD	15/07/2020
APPROVED	SBA	15/07/2020
EMPLOYER		
SCALE	1:50	A1

MANSOURAH - MASSARAH GOLD PROJECT
DETAIL DRAWING
CRUSHING
FDN LOAD DATA FOR SEC. & TER. CRUSHER HOUSE BAG FILTER

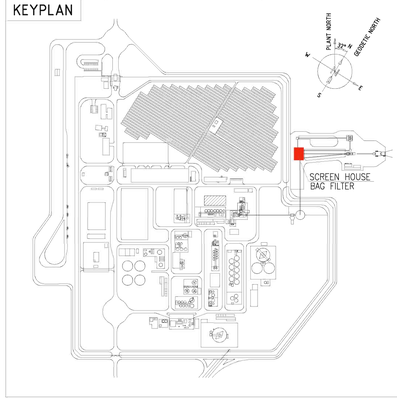
DRAWING NUMBER: MD-216-3100-EG-ME-DED-V509-2001
REV: A01

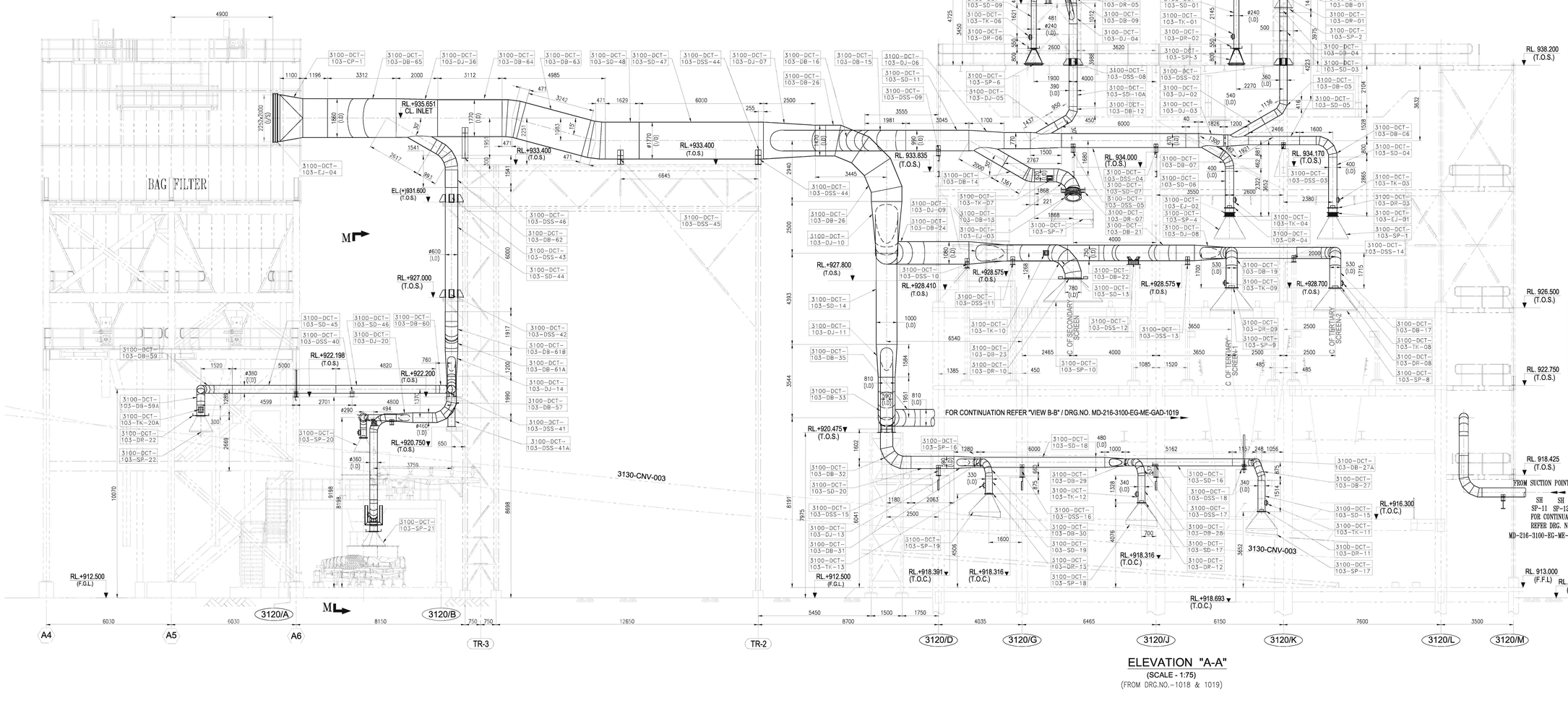
DRAWING NUMBER: MD-216-3100-EG-ME-GAD-1017 REV: A02



ELEVATION "M-M"

SCALE 1:75





ELEVATION "A-A"

(SCALE - 1:75)

(FROM DRG.NO.-1018 & 1019)

PAINTING AND PROTECTIVE COATINGS	
COATING SYSTEM	2 – 2 AS PER MD-101-SMEM-EG-ME-SPC-0007_C01 (TABLE A-1 ITEM NO. 5: DUCT)
COLOUR CODE	6032 – Signal Green as per MD-101-SMEM-EG-ME-SPC-0007_C01 (Appendix B – Equipment Colour Coding: Dust collection system)

ABBREVIATION

SP–SUCTION POINT
SD–STRAIGHT DUCT
DB–DUCT BEND
DJ–DUCT JUNCTION
CP–CHANGE PIECE
DR–MANUAL DAMPER(BAFFLE)
TK–INSPECTION COVER
DS–DUCT SUPPORT
DSS–DUCT SADDLE SUPPORT
SCH–SECONDARY CRUSHER HOUSE
DE–DUCT EXTRACTION
CP–CHANGE PIECE
TER–TERTIARY
SE–SECONDARY

NOTE:

1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. THIS DRG.TO BE READ IN CONJUNCTION WITH DRG.NO:– MD–216–3100–EG–ME–GAD–1018 TO 1023
5. DUCT MATERIAL SHALL CONFORM TO ASTM A36 /EN 10025 S275 JR OR EQUIVALENT STANDARD.
6. WELDING SHALL CONFORM TO ITB SPECIFICATION MD–101–SMEM–EG–GE–SPC–0002_C01. (GENERAL MECHANICAL FABRICATION: CARBON STEELS – AWS D1.1, AWS D1.3) OR EQUIVALENT STANDARD.
7. DUCTS DESIGNED AS PER MD–101–SMEM–EG–ME–CRT–0001 SECTION 5.3.6. AND APPROVED DESIGN–CRITERIA MD–216–3100–EG–ME–CRT–1003 A02

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3100-EG-PR-PID-V509-2002	P&ID DUST EXTRACTION – FOR SCREEN HOUSE
MD-216-3100-EG-ME-LAD-V509-2001	LAYOUT OF BAG FILTER & ID FAN
MD-2016-3270-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHED ORE STORAGE
MD-2016-3130-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING TER. CRUSHING
MD-2016-3120-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING SEC. CRUSHING
MD-2016-3110-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING PRIMARY CRUSHING
MD-316-3100-EG-ME-SDD-1001	SCHEMATIC DIAGRAM CRUSHING DUST EXTRACTION

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	27/07/2020	REVISED AS PER CUSTOMER COMMENT DATED 19.06.2020	SBR/ABW	TKS/SYR	ABD	SBA/JPD	
A01	12/05/2020	ISSUED FOR APPROVAL	TKS/SBR	ABD	ABD	SBA/JPD	

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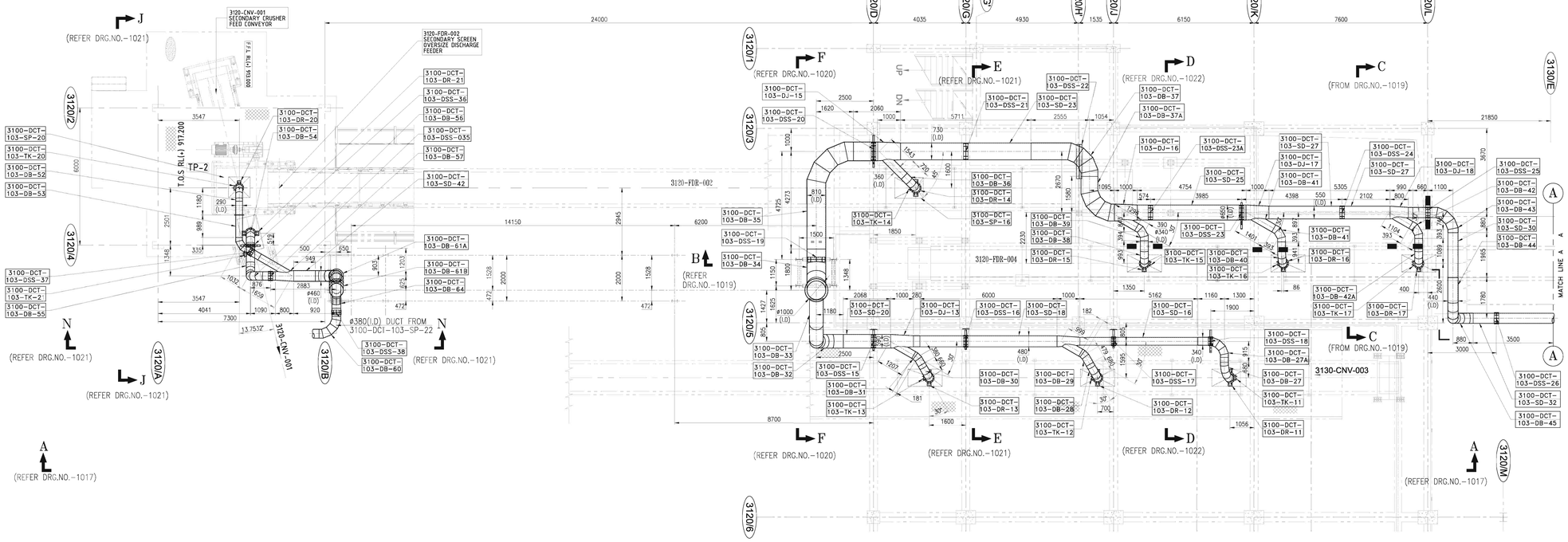
LARSEN & TOUBRO DRAWING NUMBER	019045_080_U_G_21410_001
REVISION	1
OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31005
DOCUMENT ID	OU100672966
REVISION	1

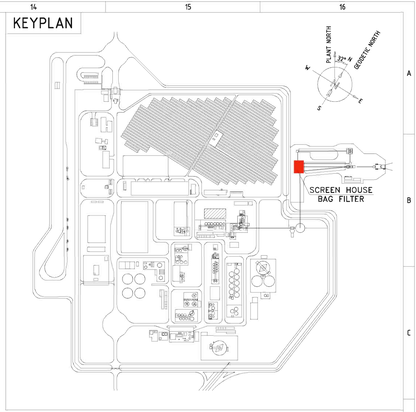
DRAWN	TKS/SBR	08/05/2020
CHECKED	ABD	11/05/2020
ENGINEER	ABD	11/05/2020
APPROVED	SBA/JPD	12/05/2020
EMPLOYER		
SCALE	1:75	A0

MANSOURAH – MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
CRUSHING
DUCT–DUST EXTRACTION SYSTEM FOR SCREEN HOUSE

DRAWING NUMBER: MD – 216 – 3100 – EG – ME – GAD – 1017
REV: A02

DRAWING NUMBER: MD-216-3100-EG-ME-GAD-1018 REV A02





PAINTING AND PROTECTIVE COATINGS	
COATING SYSTEM	2 – 2 AS PER MD-101-SMEM-EG-ME-SPC-0007_C01 (TABLE A-1 ITEM NO. 5: DUCT)
COLOUR CODE	6032 – Signal Green as per MD-101-SMEM-EG-ME-SPC-0007_C01 (Appendix B – Equipment Colour Coding: Dust collection system)

ABBREVIATION

SP–SUCTION POINT
SD–STRAIGHT DUCT
DB–DUCT BEND
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CP–CHANGE PIECE
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DSS–DUCT SADDLE SUPPORT
SCH–SECONDARY CRUSHER HOUSE

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. THIS DRG.TO BE READ IN CONJUNCTION WITH DRG.NO:- MD-216-3100-EG-ME-GAD-1017 & 1019 TO 023
5. DUCT MATERIAL SHALL CONFORM TO ASTM A36 /EN 10025 S275 JR OR EQUIVALENT STANDARD.
6. WELDING SHALL CONFORM TO ITB SPECIFICATION MD-101-SMEM-EG-GE-SPC-0002_C01. (GENERAL MECHANICAL FABRICATION: CARBON STEELS – AWS D1.1, AWS D1.3) OR EQUIVALENT STANDARD.
7. DUCTS DESIGNED AS PER MD-101-SMEM-EG-ME-CRT-0001 SECTION 5.3.6. AND APPROVED DESIGN CRITERIA MD-216-3100-EG-ME-CRT-1003 A02

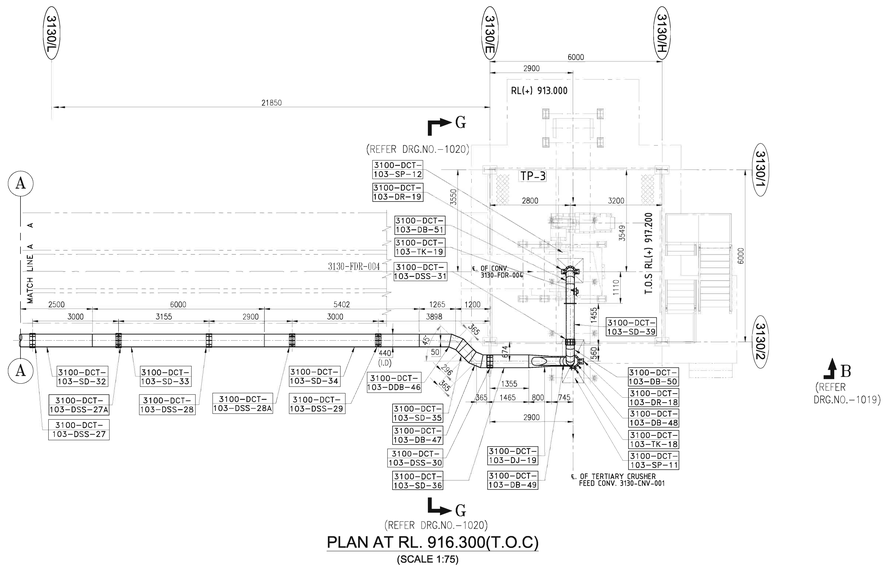

TABLE FOR SUPPORT MARKED & LOAD DATA

SL.NO.	Saddle Support Mark No.	Total Load (N Saddle(in T.)	Remarks
1	3120-DCT-103-DSS-001	1.40	
2	3100-DCT-103-DSS-002	0.70	
3	3100-DCT-103-DSS-003	1.70	
4	3100-DCT-103-DSS-004	2.70	
5	3100-DCT-103-DSS-005	3.20	
6	3100-DCT-103-DSS-006	1.10	
7	3100-DCT-103-DSS-007	0.60	
8	3100-DCT-103-DSS-008	0.50	
9	3100-DCT-103-DSS-009	10.0	
10	3120-DCT-103-DSS-010	8.0	
11	3100-DCT-103-DSS-011	4.50	
12	3120-DCT-103-DSS-012	3.0	
13	3120-DCT-103-DSS-013	3.50	
14	3120-DCT-103-DSS-014	2.20	
15	3100-DCT-103-DSS-015	2.40	
16	3100-DCT-103-DSS-016	2.0	
17	3100-DCT-103-DSS-017	2.0	
18	3100-DCT-103-DSS-018	1.0	
19	3100-DCT-103-DSS-019	8.0	
20	3100-DCT-103-DSS-020	4.2	

TABLE FOR SUPPORT MARKED & LOAD DATA

SL.NO.	Saddle Support Mark No.	Total Load (N Saddle(in T.)	Remarks
21	3100-DCT-103-DSS-021	3.10	
22	3100-DCT-103-DSS-022	3.60	
23	3100-DCT-103-DSS-023	3.0	
24	3100-DCT-103-DSS-024	1.50	
25	3100-DCT-103-DSS-025	2.20	
26	3100-DCT-103-DSS-026	1.20	
27	3100-DCT-103-DSS-027	1.0	
28	3100-DCT-103-DSS-27A	1.0	
29	3100-DCT-103-DSS-028	1.0	
30	3100-DCT-103-DSS-28A	1.0	
31	3100-DCT-103-DSS-029	1.0	
32	3100-DCT-103-DSS-030	1.50	
33	3100-DCT-103-DSS-031	1.20	
37	3100-DCT-103-DSS-035	0.70	
38	3100-DCT-103-DSS-036	0.70	
39	3100-DCT-103-DSS-037	0.80	
40	3100-DCT-103-DSS-038	0.70	
41	3100-DCT-103-DSS-039	1.40	
42	3100-DCT-103-DSS-040	1.40	
43	3100-DCT-103-DSS-041	0.80	

TABLE FOR SUPPORT MARKED & LOAD DATA

SL.NO.	Saddle Support Mark No.	Total Load (N Saddle(in T.)	Remarks
44	3100-DCT-103-DSS-41A	2.50	
45	3100-DCT-103-DSS-042	1.0	
46	3100-DCT-103-DSS-043	2.0	
47	3100-DCT-103-DSS-044	28.0	
48	3100-DCT-103-DSS-045	20.0	
49	3100-DCT-103-DSS-046	18.0	
50	3100-DCT-103-DSS-047	32.0	
51	3100-DCT-103-DSS-023A	2.10	
52	3100-DCT-103-DSS-048A	15.0	
53	3100-DCT-103-DSS-048	24.0	

A02

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3100-EG-PR-PID-V509-2002	P&ID DUST EXTRACTION – FOR SCREEN HOUSE
MD-216-3100-EG-ME-LAD-V509-2001	LAYOUT OF BAG FILTER & ID FAN
MD-2016-3210-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHED ORE STORAGE
MD-2016-3130-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING TER. CRUSHING
MD-2016-3120-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING SEC. CRUSHING
MD-2016-3110-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING PRIMARY CRUSHING
MD-316-3100-EG-ME-SDD-1001	SCHEMATIC DIAGRAM CRUSHING DUST EXTRACTION

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	27/07/2020	REVISED AS PER CUSTOMER COMMENT DATED 19.06.2020	SBR/ABW	TKS/SYR	ABD	SBA/JPD	
A01	12/05/2020	ISSUED FOR APPROVAL	TKS/SBR	ABD	ABD	SBA/JPD	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: O19045_080_U_G_21410_002 — REVISION: 1
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31006 — DOCUMENT ID: OU100672967 — REVISION: 1

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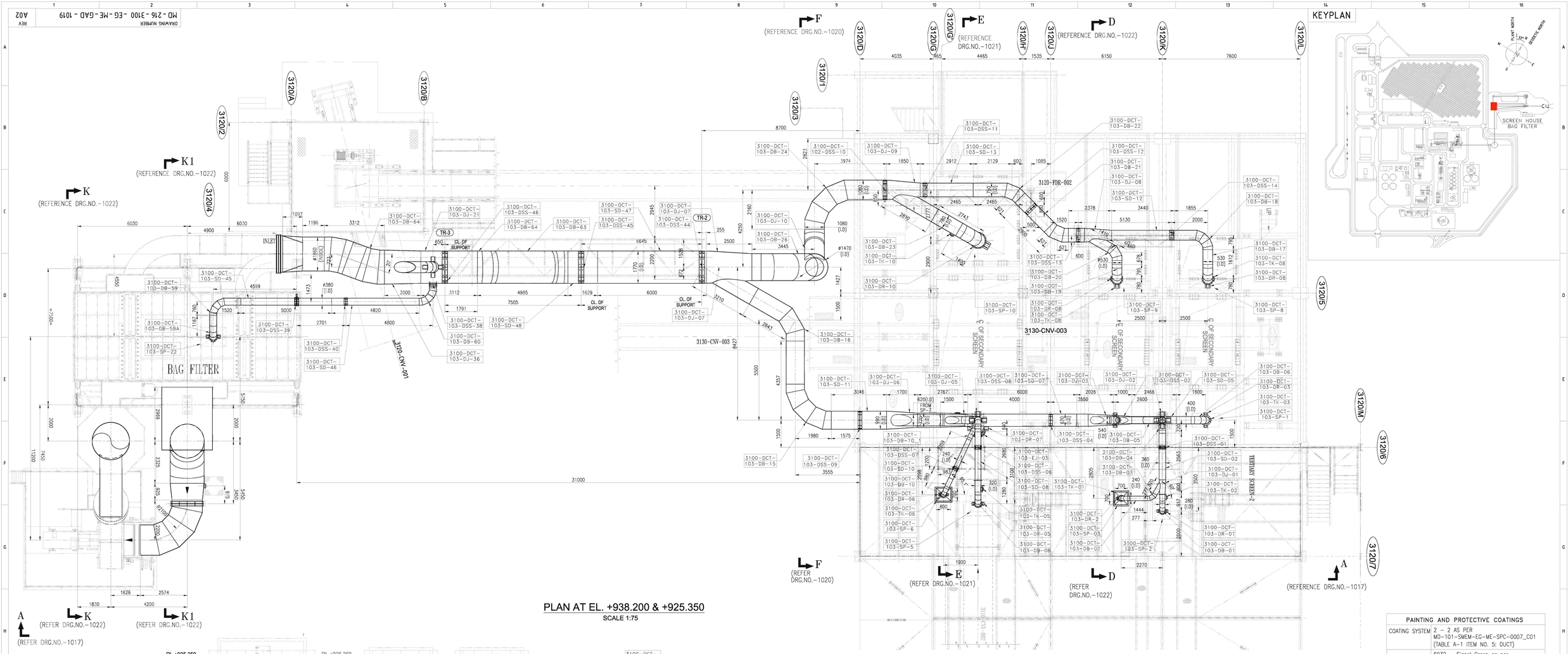
DRAWN	TKS/SBR	08/05/2020	MANSOURAH – MASSARAH GOLD PROJECT
CHECKED	ABD	11/05/2020	GENERAL ARRANGEMENT
ENGINEER	ABD	11/05/2020	CRUSHING
APPROVED	SBA/JPD	12/05/2020	DUCT-DUST EXTRACTION SYSTEM FOR SCREEN HOUSE
EMPLOYER			
SCALE	1:75	A0	DRAWING NUMBER: MD-216-3100-EG-ME-GAD-1018 — REV: A02

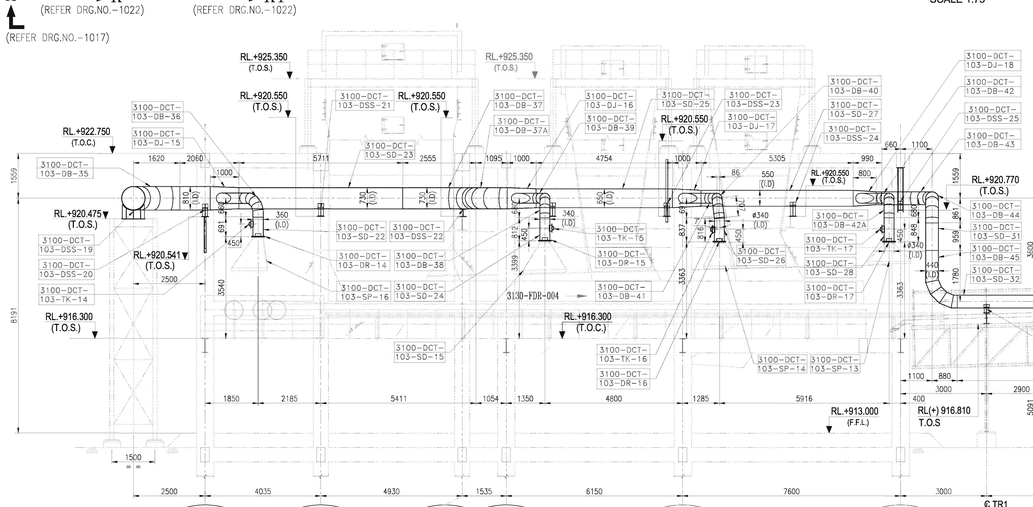
DRAWING NUMBER: MD-216-3100-EG-ME-GAD-1019 REV: A02

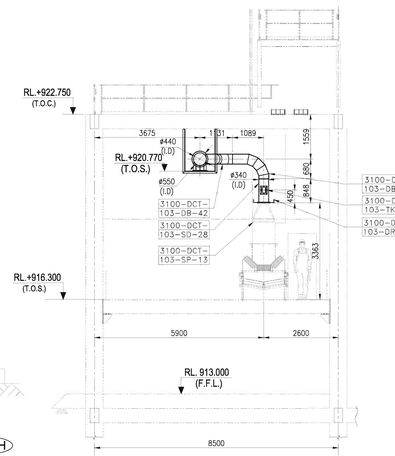


PAINTING AND PROTECTIVE COATINGS	
COATING SYSTEM	2 – 2 AS PER MD-101-SMEM-EG-ME-SPC-0007_C01 (TABLE A-1 (ITEM NO. 5: DUCT)
COLOUR CODE	6032 – Signal Green as per MD-101-SMEM-EG-ME-SPC-0007_C01 (Appendix B – Equipment Colour Coding: Dust collection system)

ABBREVIATION

SP-SUCTION POINT
SD-STRAIGHT DUCT
DB-DUCT BEND
DJ-DUCT JUNCTION
CP-CHANGE PIECE
DR-MANUAL DAMPER(BAFFLE)
TK-INSPECTION COVER
DS-DUCT SUPPORT
DSS-DUCT SADDLE SUPPORT
SCH-SECONDARY CRUSHER HOUSE
DE-DUST EXTRACTION
CP-CHANGE PIECE
TER-TERTIARY
SE-SECONDARY





NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. THIS DRG.TO BE READ IN CONJUNCTION WITH DRG.NO:- MD-216-3100-EG-ME-GAD-1017 ,1018 & 1020 TO 1023
5. DUCT MATERIAL SHALL CONFORM TO ASTM A36 /EN 10025 S275 JR OR EQUIVALENT STANDARD.
6. WELDING SHALL CONFORM TO ITB SPECIFICATION MD-101-SMEM-EG-GE-SPC-0002_C01. (GENERAL MECHANICAL FABRICATION: CARBON STEELS – AWS D1.1, AWS D1.3) OR EQUIVALENT STANDARD.
7. DUCTS DESIGNED AS PER MD-101-SMEM-EG-ME-CRT-0001 SECTION 5.3.6. AND APPROVED DESIGN CRITERIA MD-216-3100-EG-ME-CRT-1003 A02

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3100-EG-PR-PID-V509-2002	P&ID DUST EXTRACTION – FOR SCREEN HOUSE
MD-216-3100-EG-ME-LAD-V509-2001	LAYOUT OF BAG FILTER & ID FAN
MD-316-3100-EG-ME-SDD-1001	PROCESS FLOW DIAGRAM CRUSHED ORE STORAGE
MD-2016-3130-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING TER. CRUSHING
MD-2016-3120-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING SEC. CRUSHING
MD-2016-3110-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING PRIMARY CRUSHING
MD-316-3100-EG-ME-SDD-1001	SCHEMATIC DIAGRAM CRUSHING DUST EXTRACTION

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	27/07/2020	REVISED AS PER CUSTOMER COMMENT DATED 19.06.2020	SBD/ABW	TKS/SYR	ABD	SBA/JPD	
A01	12/05/2020	ISSUED FOR APPROVAL	TKS/SBR	ABD	ABD	SBA/JPD	

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LARSEN & TOUBRO DRAWING NUMBER	019045_080_U_G_21410_003
REVISION	1
OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31007
DOCUMENT ID	OU100672968
REVISION	1
DRAWN	TKS/SBR 08/05/2020
CHECKED	ABD 11/05/2020
ENGINEER	ABD 11/05/2020
APPROVED	SBA/JPD 12/05/2020
EMPLOYER	
SCALE	1:75
	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
CRUSHING
DUCT-DUST EXTRACTION SYSTEM FOR SCREEN HOUSE

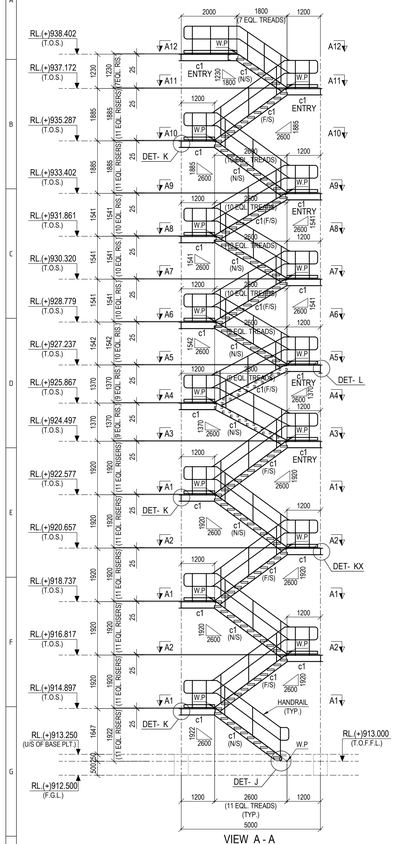
DRAWING NUMBER: MD - 216 - 3100 - EG - ME - GAD - 1019
REV: A02

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

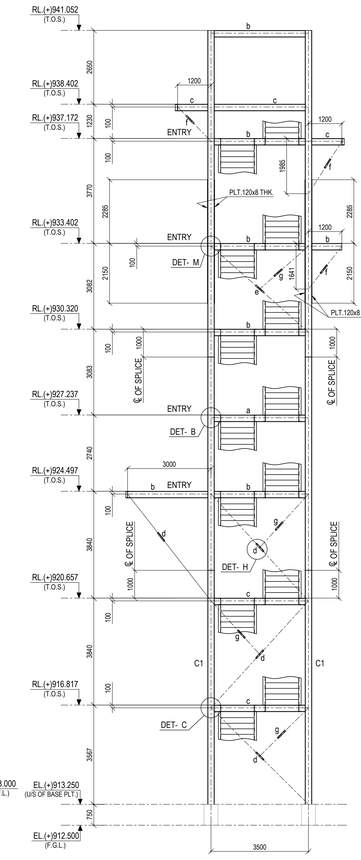
Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	03-09-2020	14-10-2020	31-10-2023				Date	03-09-2020	14-10-2020	31-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 2 **STRUCTURAL GENERAL ARRANGEMENT DRAWINGS CONTAINING BASE PLATE DETAIL**

DRAWING NUMBER: MD-216-3100-EG-ST-GAD-2024 REV A01



VIEW A - A
(SCALE- 1:75)



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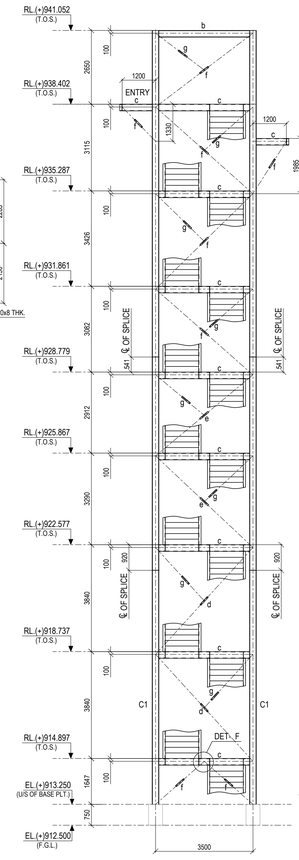

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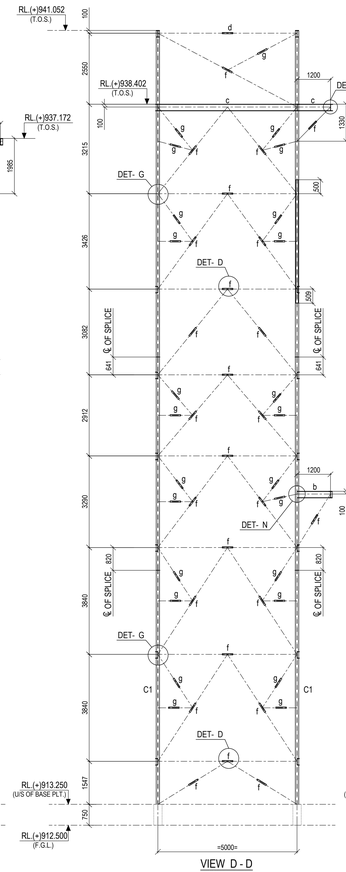

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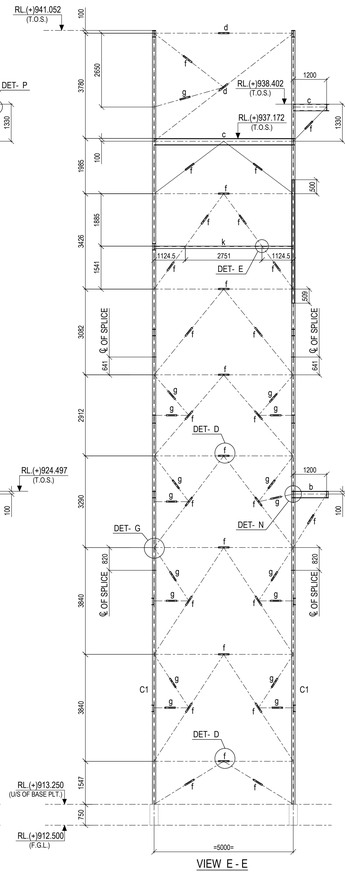

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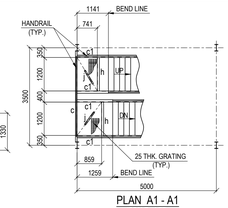

PLAN A1 - A1
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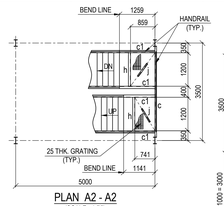

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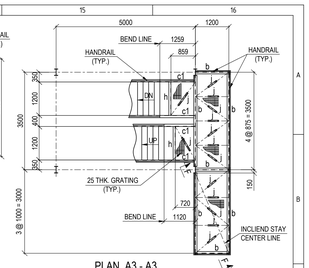

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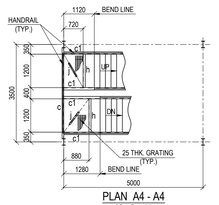

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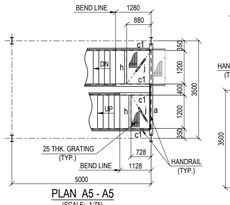

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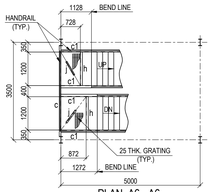

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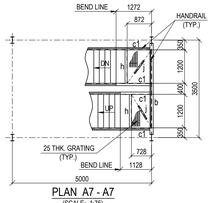

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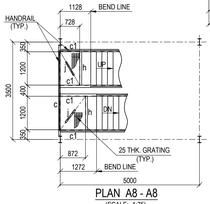

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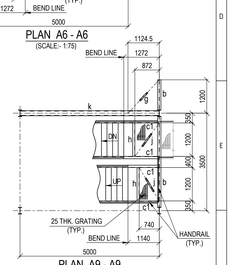

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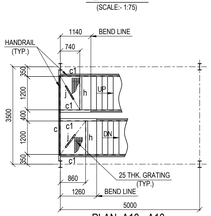

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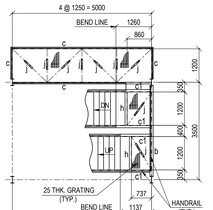

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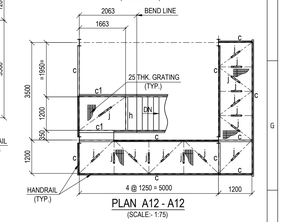

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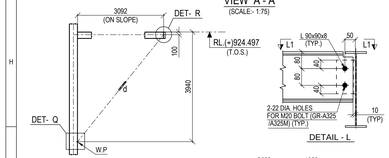

VIEW F - F
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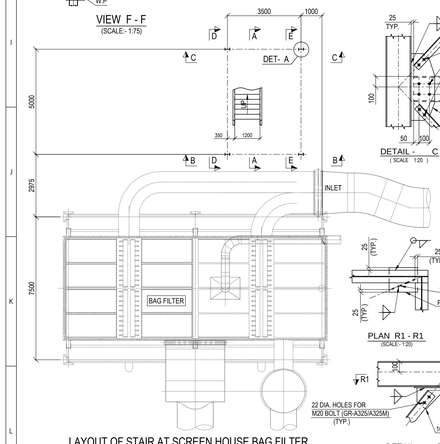
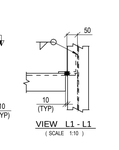
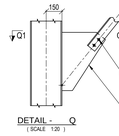
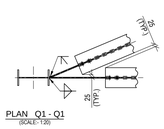
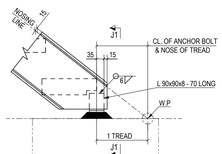


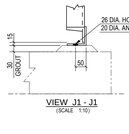
LAYOUT OF STAIR AT SCREEN HOUSE BAG FILTER
(SCALE- 1:100)

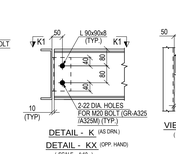


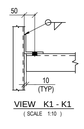


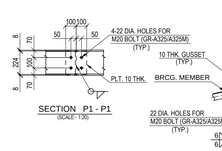


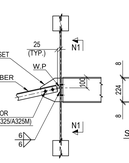


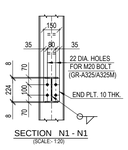


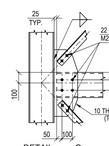


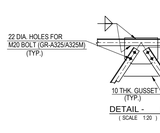


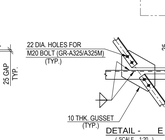


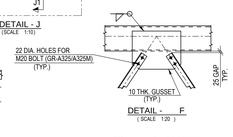


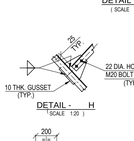


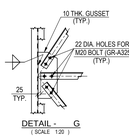


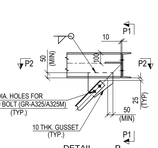


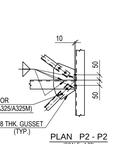


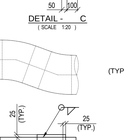


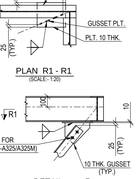


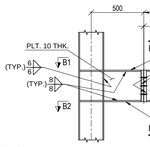


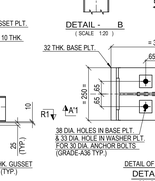


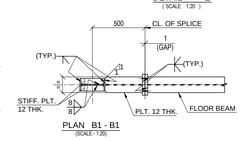


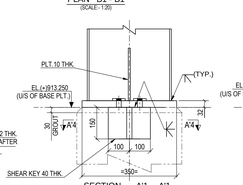


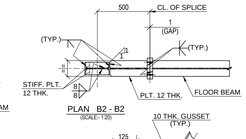


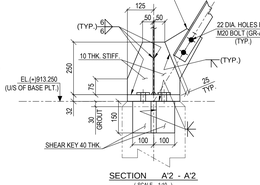


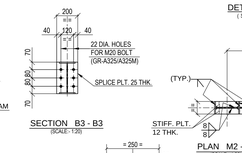


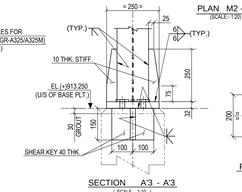


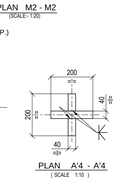


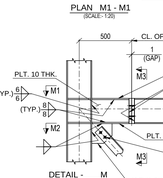












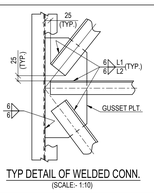
LEGENDS:-

TOS - TOP OF STEEL
U/S - UNDER SIDE
CL. - CENTER LINE
RL - REDUCED LEVEL
T.O.F.F.L. - TOP OF FINISHED FLOOR LEVEL
FGL - FINISHED GROUND LEVEL
F/S - FAR SIDE
N/S - NEAR SIDE
DN. - DOWN
W.P. WORKING POINT
GR. - GRATING
TYP. - TYPICAL
PLT. - PLATE
THK. - THICKNESS
STIFF. - STIFFNER
CONN. - CONNECTION

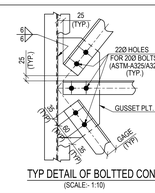
SCHEDULE OF MEMBERS

MEMBER MKD.	MEMBER	SECTON	FACTORED AXIAL FORCE(KN)	FACTORED SHEAR FORCE(KN)	NO. OF BOLTS.	WELD LENGTH (mm) (HEEL SIDE) L1	WELD LENGTH (mm) (TOE SIDE) L2	REMARKS
C1	IPE 300	I	-	-	-	-	-	MATERIAL GRADE:S275 JR
a	IPE300	I	10.0	70.0	6	-	-	
b	IPE240	I	70.0	50.0	4	-	-	
c	UPN240	C	150.0	50.0	4	-	-	
c1	UPN240	C	20.0	50.0	4	-	-	
d	L - 90X90X8	⌐	210.0	-	3	-	-	
e	L - 75X75X8	⌐	140.0	-	2	-	-	
f	L - 65X65X6	⌐	130.0	-	2	-	-	
g	L - 65X65X6	⌐	-	-	2	-	-	
h	UPN200	C	-	-	4	-	-	
j	L - 50X50X6	⌐	-	-	2	-	-	
k	UPN240	C	150.0	50.0	4	-	-	

ALL GUSSET PLATE ARE 10mm THK. UNLESS OTHERWISE NOTED.
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.



TYP DETAIL OF WELDED CONN.
(SCALE- 1:10)



TYP DETAIL OF BOLTTED CONN.
(SCALE- 1:10)

NOTES:

1. ALL DIMENSIONS & CO-ORDINATES ARE IN MILLIMETER AND LEVELS ARE IN METER UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36.
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
4. ALL WELDS SHALL BE OF SHOP QUALITY & TO BE TESTED AS PER QUALITY ASSURANCE PLAN.
5. ALL PERMANENT STRUCTURAL BOLTS CONFORM TO ASTM A 325M.
6. PAINTING ON STRUCTURES WILL BE AS PER DESIGN BASIS REPORT (DBR).
7. FOR SPLICE DETAIL REFER DRAWING NO. (MD-216-1000-EG-ST-DED-1011)

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-ST-CRT-1001	CIVIL,STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA	A01	04/09/2020	ISSUED FOR APPROVAL	HSK	SBY	RPK	PRI	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: O90645_080_S_G_27013_005 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31051 — DOCUMENT ID: QU100692089 — REVISION: 0

DRAWN	HSK	31/08/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SBY	01/09/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	RPK	02/09/2020	SCREEN BUILDING BAG FILTER
APPROVED	PRI	04/09/2020	PLAN, ELEVATION AND DETAIL OF STAIR
EMPLOYER			
SCALE 1 : 10, 20, 75 & 100		A0	DRAWING NUMBER: MD-216-3100-EG-ST-GAD-2024 — REV: A01

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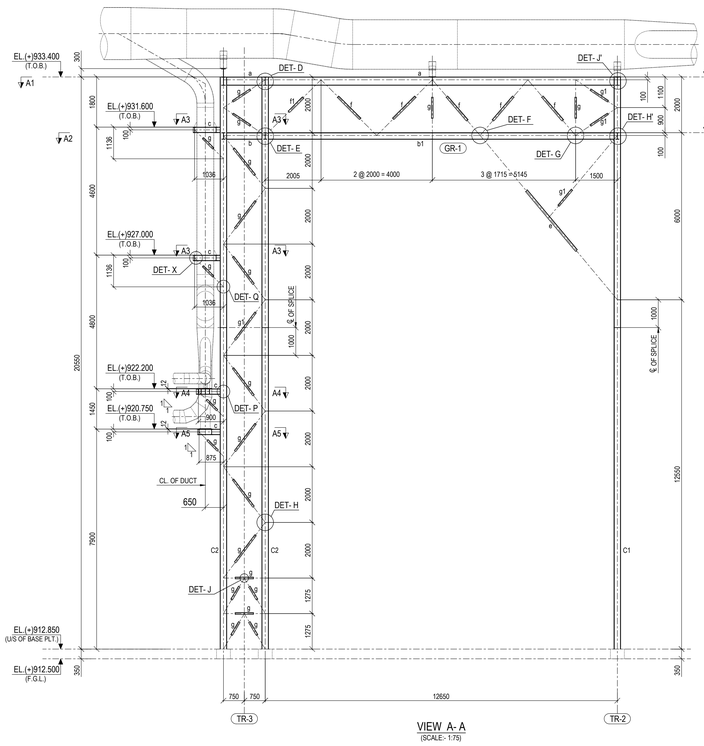
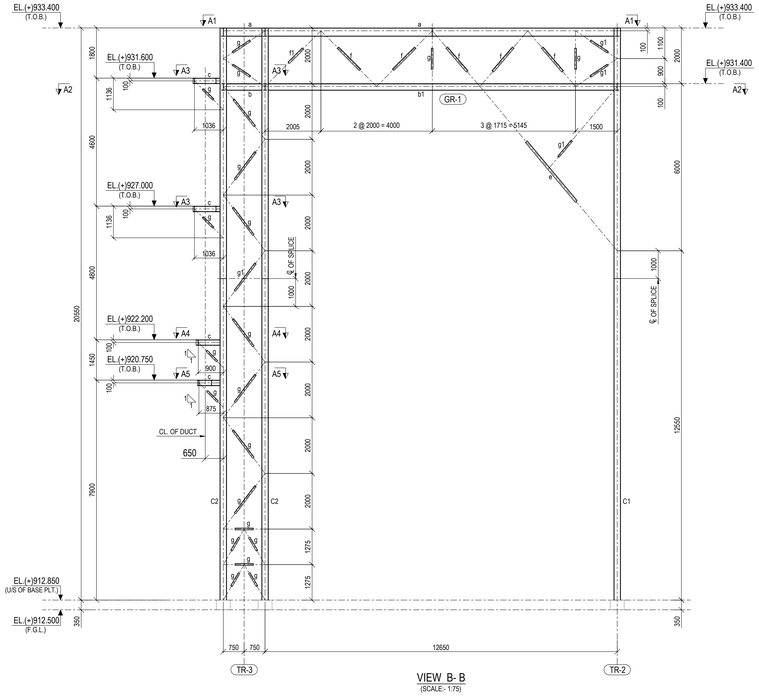
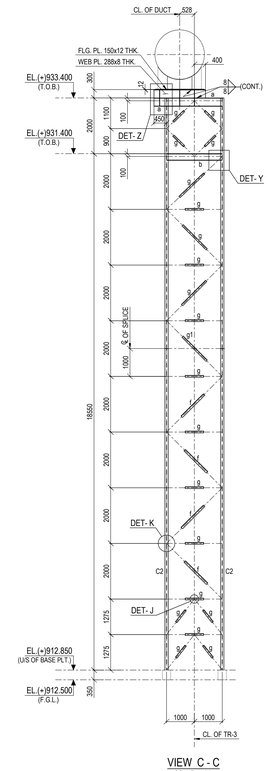
DRAWING NUMBER: MD-216-3120-EG-ST-GAD-2022 | REV: A01



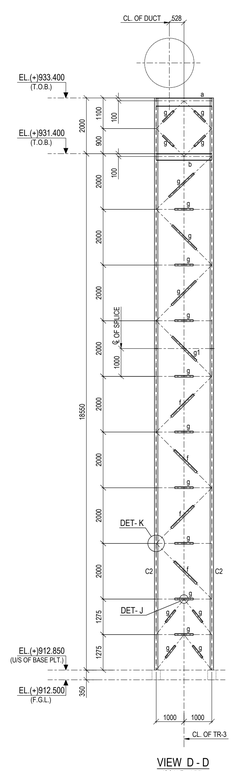
VIEW A- A
(SCALE:- 1:75)



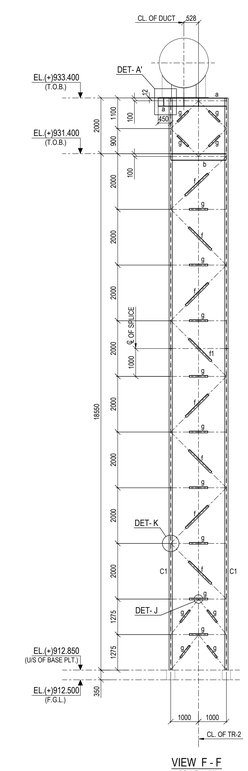
VIEW B- B
(SCALE:- 1:75)



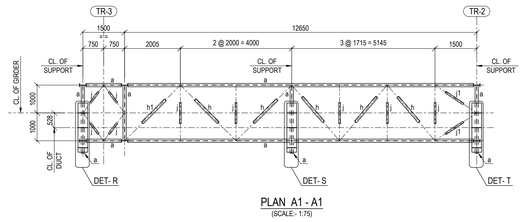
VIEW C - C
(SCALE:- 1:75)



VIEW D - D
(SCALE:- 1:75)



VIEW F - F
(SCALE:- 1:75)



PLAN A1 - A1
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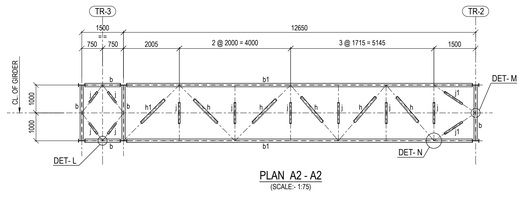

PLAN A2 - A2
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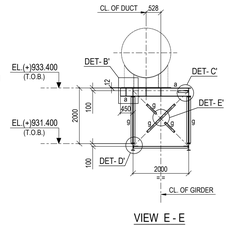
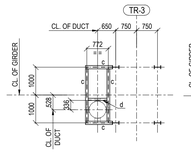


VIEW E - E
(SCALE:- 1:75)



PLAN A3 - A3
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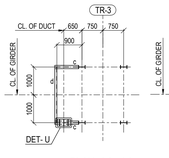
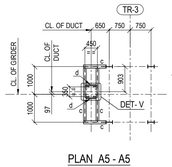


PLAN A4 - A4
(SCALE:- 1:75)



PLAN A5 - A5
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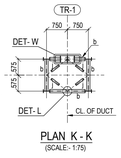

PLAN K - K
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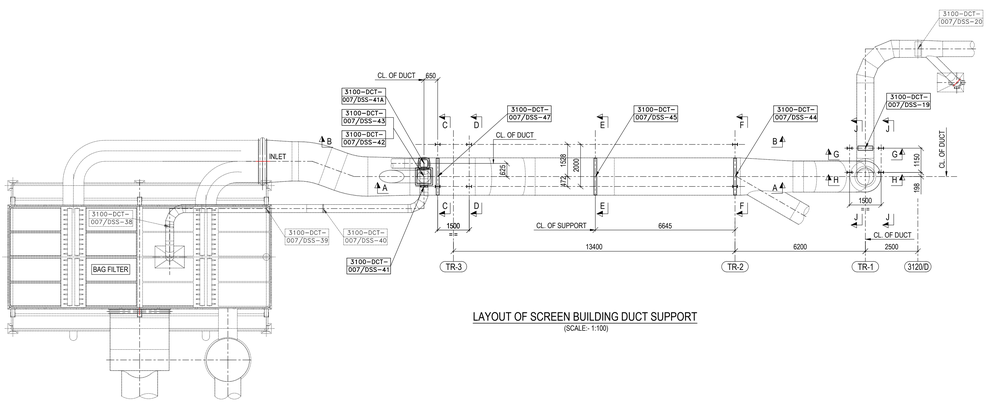
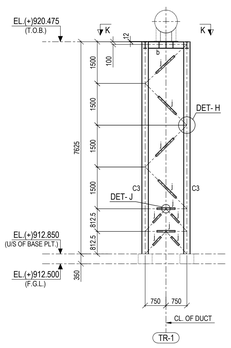
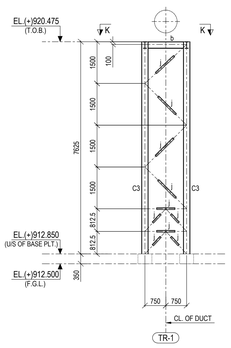


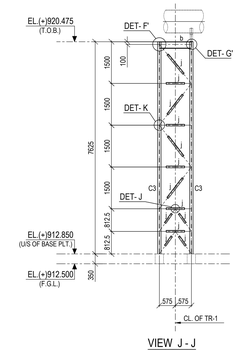
LAYOUT OF SCREEN BUILDING DUCT SUPPORT
(SCALE:- 1:100)



VIEW G - G
(SCALE:- 1:75)



VIEW H - H
(SCALE:- 1:75)

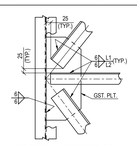


VIEW J - J
(SCALE:- 1:75)

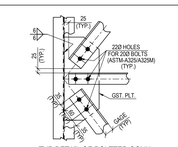
SCHEDULE OF MEMBERS

MEMBER MKD.	MEMBER	SECTON	FACTORED AXIAL FORCE(KN)	FACTORED SHEAR FORCE(KN)	NO. OF BOLTS.	WELD LENGTH (mm) (HEEL SIDE) L1	WELD LENGTH (mm) (TOE SIDE) L2	REMARKS
C1	IPE 360	I	-	-	-	-	-	MATERIAL GRADE:S275 JR
C2	IPE 300	I	-	-	-	-	-	
C3	IPE 200	I	-	-	-	-	-	
a	IPE 300	I	170	150	-	-	-	
b	IPE 240	I	200	90	-	-	-	
b1	1/WEB PLT. 140x10 THK. 2/FLG. PLT. 220x8 THK.	I	200	90	-	-	-	
c	IPE 200	I	-	-	-	-	-	
d	UPN 200	C	-	-	-	-	-	
e	L - 90X90X8	⊢	-	-	2	-	-	
f	L - 75X75X8	⊢	170	-	-	150	75	
f1	L - 75X75X8	⊢	150	-	3	-	-	
g	L - 65X65X6	⊢	190	-	-	150	75	
g1	L - 65X65X6	⊢	110	-	2	-	-	
h	L - 90X90X8	⌐	40	-	-	150	75	
h1	L - 90X90X8	⌐	40	-	2	-	-	
j	L - 65X65X6	⌐	30	-	-	150	75	
j1	L - 65X65X6	⌐	20	-	2	-	-	

ALL GUSSET PLATE ARE 8mm THK. UNLESS OTHERWISE NOTED.
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.



TYP DETAIL OF WELDED CONN.
(SCALE:- 1:10)



TYP DETAIL OF BOLTTED CONN.
(SCALE:- 1:10)

NOTES:

1. ALL DIMENSIONS & CO-ORDINATES ARE IN MILLIMETER AND LEVELS ARE IN METER UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36.
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
4. ALL WELDS SHALL BE OF SHOP QUALITY & TO BE TESTED AS PER QUALITY ASSURANCE PLAN.
5. ALL PERMANENT STRUCTURAL BOLTS CONFORM TO ASTM A 325M.
6. PAINTING ON STRUCTURES WILL BE AS PER DESIGN BASIS REPORT (DBR).
7. FOR SPLICE DETAIL REFER DRAWING NO. (MD-216-1000-EG-ST-DED-1011)

LEGENDS:-

TOB - TOP OF BEAM
U/S - UNDER SIDE
CL. - CENTER LINE
TYP. - TYPICAL
PLT. - PLATE
THK. - THICKNESS
FGL - FINISHED GROUND LEVEL
B/S - BOTH SIDE
GST. - GUSSET
STIFF. - STIFFNER
CONT. - CONTINUE
CONN. - CONNECTION

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-ST-CRT-1001	CIVIL,STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA	A01	10/08/2020	ISSUED FOR REFERENCE	HSK	SBY	RPK	PRI	

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LARSEN & TOUBRO DRAWING NUMBER	019045_080_S_G_27013_003		REVISION 0
OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31284	DOCUMENT ID OU100692111	REVISION 0
DRAWN	HSK	01/08/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SBY	03/08/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	RPK	04/08/2020	SCREEN BUILDING
APPROVED	PRI	06/08/2020	PLAN, ELEVATION AND DETAIL OF DEDUSTING DUCT SUPPORT
EMPLOYER			
SCALE 1 : 10, 75 & 100	A0	DRAWING NUMBER MD-216-3120-EG-ST-GAD-2022	REV A01